

LLM-as-a-Judge Rubric Types for Open-Ended Evaluation (2022–2025)

Large language models have been used as **evaluators** (“LLM-as-a-judge”) to grade or compare open-ended responses. Recent academic work highlights several major rubric types for these model-based evaluations. Below, we summarize the key rubric formats – binary judgments, point-scale ratings, structured multi-criteria rubrics, and ranking comparisons – along with their sources, structures, evaluation targets, and any special scoring protocols.

Binary Correctness Rubrics

These rubrics reduce evaluation to a **binary decision** (e.g. *correct vs. incorrect* or *pass vs. fail*). They are used when an open-ended answer can objectively be judged as right or wrong:

- **TruthfulQA (Lin et al., 2022)** – *Source*: NeurIPS 2022 benchmark on truthfulness. *Structure*: **Binary label** for each answer (“truthful” or “false”). *Target*: Factual accuracy in open-ended question answering. *Protocol*: Human evaluators labeled each answer as truthful if it contained no falsehoods, or false if it was misleading/untrue. The best GPT-3 model was truthful on only ~58% of questions under this rubric (versus 94% for humans) ¹, reflecting the difficulty of binary truthfulness evaluation.
- **Pass/Fail Factuality Checks** – *Source*: Hallucination detection evaluations (e.g. HaluEval/HaluBench 2023). *Structure*: **Pass/Fail** outcome indicating if an answer is entirely accurate. *Target*: Factual consistency of generated answers against a source or ground truth. *Protocol*: The LLM judge marks an answer “Pass” if it contains **no factual errors** and “Fail” if any part is unfaithful to the context. For example, FlowAI’s open evaluator model supports a simple pass/fail rubric for error detection tasks ². In one setup, answers to knowledge queries were labeled *pass* only if they aligned with reference facts; hallucinated yet plausible answers were flagged *fail* ³. Evaluation is then as straightforward as computing accuracy (% of passes) ⁴.
- **Binary Code Correctness** – *Source*: Code generation evals (e.g. OpenAI “HumanEval”). *Structure*: **Correct/Incorrect** based on test outcomes. *Target*: Code solutions to programming tasks. *Protocol*: While often automated, some LLM-judges are tasked with predicting if code would pass all unit tests (essentially a binary rubric). For instance, an LLM can be prompted to read a code submission and respond “Correct” if it meets the problem requirements or “Incorrect” otherwise. This pass/fail rubric mirrors how code is graded against an expected output (0 or 1 score). Special instructions usually encourage the judge to explain any errors when labeling a solution incorrect (providing feedback), though the final score remains binary.

Note: Binary rubrics work best for *objective* questions (e.g. factual QA, math, code), where a clear ground truth exists. They are less suited to subjective tasks, which require more nuanced scales.

Point-Scale Rubrics (Likert or Numeric Scores)

Point-scale rubrics assign a **numerical rating** to a single answer along one or more dimensions. Commonly, this is a Likert-type scale (e.g. 1 to 5). This approach provides more gradation than binary judgment:

- **Likert Quality Ratings (Summarization & QA)** – *Source:* Traditional human eval protocols (e.g. SummEval, FRANK 2021). *Structure:* **5-point Likert scale** (1 = very poor, 5 = excellent) on defined criteria. *Target:* Overall quality or specific aspects (coherence, relevance, etc.) of a generated text. *Protocol:* Evaluators rate an answer or summary on one dimension at a time. For example, a summarization study had crowd workers score summaries from 1 to 5 across four key dimensions: *coherence, consistency, fluency, and relevance* ⁵. Similarly, in a medical QA setting, clinicians rated each answer’s *correctness* and *usefulness* on a 5-point scale ⁶. LLM judges can mimic this by outputting a rating with an explanation if prompted appropriately. The key instruction is to define what each score means (e.g. 1 = completely irrelevant, 5 = perfectly accurate).
- **Holistic Single-Score (GPT-4 Scoring)** – *Source:* OpenAI and Microsoft research (G-Eval by Liu et al., 2023). *Structure:* **Numeric scale** (often 1–5 or 1–10) for an overall judgment. *Target:* General quality of a response (combining factors like correctness, clarity, completeness). *Protocol:* The LLM judge is given a prompt describing the scoring criteria and asked to produce a single score. For instance, G-Eval introduces a prompt where GPT-4 is instructed: “Evaluate the answer’s quality on a scale from 1 to 5, considering coherence, relevance, and correctness (1=Very Poor, 5=Excellent)”. The model “fills in” the score in a form along with justification ⁷. Experiments showed GPT-4 can leverage its internal knowledge of language quality to reliably differentiate fluent, coherent text from low-quality text when scoring answers ⁸. Special protocols like **chain-of-thought (CoT)** prompting are sometimes used: GPT-4 is asked to reason step-by-step about the answer before deciding on a score, which can improve consistency ⁹.
- **Custom Scales and Calibration** – *Source:* Evaluation toolkits (e.g. FlowAI’s FlowJudge 2024). *Structure:* **Various point scales** tailored to the task (e.g. 3-point or 7-point scales). *Target:* When a fine-grained or coarse judgment is preferred. *Protocol:* Some evaluators experiment with different scale lengths. For example, FlowJudge was trained on datasets with **3-level** (“low/medium/high”) and **5-level** Likert ratings in addition to binary labels ¹⁰. Using fewer levels can force a clearer distinction in quality, whereas more levels allow nuance. In practice, 5-point scales are most common for language tasks due to their balance of simplicity and nuance. Prompting must clarify the scale (what each number means) to yield reliable outputs. Notably, recent studies found that *LLMs can be inconsistent if asked for a raw numeric rating without guidance*, sometimes exhibiting biases in rating distribution ¹¹. Thus, evaluators often include anchor descriptions for each scale point or prefer comparative judgments for subjective qualities (see below).

Structured Criteria Rubrics (Multi-Dimensional)

Structured rubrics evaluate **multiple dimensions** of an answer separately. Instead of one overall score, the judge considers several criteria (e.g. correctness, reasoning, clarity, style) and provides a structured assessment. This yields a more comprehensive profile of strengths and weaknesses:

- **Fine-Grained Skill Rubrics (FLASK, 2024)** – Source: Ye et al. (2024), *Fine-grained Language Model Evaluation based on Alignment Skill*. Structure: **Multiple scoring dimensions** (each on a point scale). Target: Diverse “skills” in generated text (logic, knowledge, etc.). Protocol: The FLASK framework defines a broad set of evaluation skills – e.g. *Logical Robustness, Logical Correctness, Efficiency, Commonsense Understanding, Factuality, Readability*, and more ¹² ¹³. An LLM judge is prompted to rate *each skill* for a given answer (often on a 1–5 or 1–10 scale). The result is a vector of scores, one per criterion. Special instructions ensure the judge focuses only on the specified aspect when scoring that aspect (to avoid one criterion’s flaws biasing another). This structured approach captures where a model excels or fails; for example, a text might score high on **fluency** but low on **factuality**, flagging a hallucination issue. FLASK showed high correlation between such model-generated multi-aspect scores and human evaluations ¹⁴.
- **Composite Criteria for Q&A (ScienceQA Rubrics)** – Source: Pride et al. (2023); Babaei Giglou et al. (2024). Structure: **Several named criteria**, each with its own rating. Target: Long-form answers in scientific Q&A. Protocol: In a science question-answer evaluation, an initial rubric defined three core criteria – **Comprehensiveness, Trustworthiness, and Utility** of the answer ¹⁵. Later work expanded this to **nine specific rubrics** covering granular aspects (e.g. accuracy of content, depth of explanation, clarity of language, presence of uncertainty qualifiers, etc.) ¹⁶. An LLM judge rates each aspect on a 5-point scale, following detailed instructions for what constitutes a high vs. low score on that dimension. For example, *Comprehensiveness* might be rated 5 if the answer addresses **all parts of the question thoroughly**, or 1 if it misses major points. The **scoring protocol** often provides example good/bad answers for each criterion to calibrate the model. By combining the rubric scores, researchers can compute an overall grade or analyze which aspect caused a model’s answer to fail. This method was used to evaluate scientific QA models without relying on exact-match answers, by focusing on quality facets ¹⁷.
- **Form-Filled Evaluator Prompt (G-Eval, 2023)** – Source: Liu et al. (2023), *G-Eval: NLG Evaluation using GPT-4*. Structure: **Pre-defined form** with multiple fields (criteria) to fill. Target: Summary and dialogue outputs (evaluating multiple qualities). Protocol: G-Eval uses GPT-4 with an **evaluation form** prompting approach ⁹. The prompt begins with a description of the task and a list of evaluation criteria (e.g. *Coherence, Conciseness, Relevance, Grammar* for a summary) including the rating scale for each ⁷. GPT-4 is instructed to first produce a *chain-of-thought reasoning* assessing the text on each criterion, then output the form with scores for each aspect and an optional explanation. For instance, for *Coherence (1–5)* the model might internally consider if the summary’s ideas logically flow, then assign a score. *Special instructions*: The model must not skip any field; it uses a consistent format (e.g. “Coherence: 4; Justification: ...”). This structured rubric with CoT improved alignment to human judgments (GPT-4’s aspect scores had higher correlation with human ratings than a raw single score) ¹⁸ ¹⁹. G-Eval demonstrated that with careful prompt design, LLMs can *simultaneously* evaluate multiple facets and even give interpretable rationales for each facet.

- **Conversation Quality Rubrics (RUBICON, 2024)** – *Source:* Biyani et al. (2024), Microsoft Research. *Structure:* **Rubric assertions** + Likert agreement. *Target:* Multi-turn human–AI dialogues (user satisfaction and quality). *Protocol:* The RUBICON approach automatically **generates a set of rubric statements** about a conversation (e.g. “The assistant’s responses addressed the user’s needs thoroughly”, “The assistant’s tone was appropriate and engaging”). These statements cover various quality aspects (naturalness, engagement, correctness, etc.) ²⁰ ²¹ . An instruction-tuned LLM then evaluates each statement on a 5-point Likert scale indicating how much it agrees (5 = strongly agree, 1 = strongly disagree) given the conversation ¹¹ . The multiple ratings are aggregated into an overall **Net Satisfaction (NetSAT)** score for the dialogue ²² – effectively the difference between satisfied vs. dissatisfied aspects. *Special scoring protocol:* If the LLM’s scores on certain assertions are very low (indicating major problems), the conversation might be classified as “negative” overall once a threshold is exceeded ²³ . By breaking evaluation into many small judgments, this rubric reduces subjectivity: the model focuses on one quality at a time. Biyani et al. report that using a fixed Likert scale for each assertion was more consistent than trying to have the LLM produce a continuous score directly ²⁴ . This illustrates how structured criteria can yield a more reliable and interpretable evaluation of open-ended interactions.

Note: Multi-dimensional rubrics demand careful prompt design to avoid confusion between criteria. They are powerful for in-depth evaluation (identifying *why* a response is good or bad in different ways). However, using many criteria can be time-consuming or costlier (each extra judgment is additional work for the model). Some studies therefore select a minimal optimal subset of rubric aspects to balance thoroughness and efficiency ²⁵ ²⁶ .

Ranking Comparison Rubrics

Ranking-based rubrics involve **comparing multiple responses** to determine which is better. Typically, this is done pairwise (two at a time), asking the LLM-judge to pick a winner (or declare a tie). This approach aligns closely with how human preference tests and RLHF training are conducted:

- **Pairwise Preference Evaluation** – *Source:* Numerous LLM eval studies in 2023 (e.g. Wang et al. 2023; Chiang & Lee 2023; Zheng et al. 2023). *Structure:* **A vs. B comparison** (with optional tie). *Target:* Any open-ended task where quality is subjective or multifaceted (chat, storytelling, etc.). *Protocol:* The judge is given a question or prompt **plus two candidate answers**, and must decide which answer is better overall according to a given rubric or criteria list. Early “LLM-as-judge” work primarily relied on pairwise comparisons to mimic human preference judgments ²⁷ . For example, in the **Chatbot Arena** benchmark, GPT-4 was prompted with the entire conversation context and two different model responses, then asked: “Which assistant’s response is more helpful, correct, and coherent?”. The LLM outputs either “Assistant A is better”, “Assistant B is better”, or “They are equally good,” often along with an explanation. Zheng et al. (2023) found that a strong LLM judge (GPT-4) could match human pairwise preferences with about **80% agreement** on open-ended questions ²⁸ ²⁹ . Special instructions are used to reduce bias – e.g. the prompt might hide which model wrote which answer (to avoid favoring itself), and require a justification before the final choice (to encourage careful deliberation). Pairwise rubrics have been shown to yield more stable rankings and better human alignment than direct scoring for subjective qualities ³⁰ . This is because it’s easier (for both humans and models) to say “out of these two, which is better” than to assign absolute scores in isolation ³⁰ .

- **Rank-Ordering and Elo** – *Source*: Crowdsourced leaderboards (LMSys, 2023). *Structure*: **Comparative ranking** derived from many pairwise battles. *Target*: Overall ranking of multiple models or multiple answers. *Protocol*: Although the LLM judge only compares two responses at a time, these pairwise wins/losses can be aggregated into a global ranking. For instance, the LMSys Chatbot Arena assigns Elo ratings to models based on win/loss outcomes from many GPT-4 vs. GPT-4 matchups ²⁸. In academia, Gao et al. (2023) similarly had models compare answers in round-robin tournaments to see which model is strongest on a set of queries. *Special considerations*: Directly asking an LLM to **rank a list** of 3+ responses in one go (so-called *listwise evaluation*) is less common – it can lead to inconsistent judgments (violating transitivity) ³¹. For example, an LLM might rate response A and B both highly when seen separately, but when directly ranking A, B, C together it might produce a contradictory order. Thus, most ranking rubrics stick to repeated pairwise comparisons and then use algorithms to derive a final ordering ³² ³¹. If ties are allowed, those are treated as equal scores or resolved via additional criteria. The **scoring protocol** for pairwise prompts often emphasizes specific criteria: e.g. “*Consider factual accuracy first, then completeness, then style. Decide which answer is better overall.*” This helps the model focus on important aspects when two answers each have different strengths.

- **Human Preference Emulation (RLHF)** – *Source*: InstructGPT (Ouyang et al. 2022), Anthropic HHH model (Bai et al. 2022). *Structure*: **Preference ranking** (often pairwise) used to train reward models. *Target*: Chat assistant helpfulness, harmlessness, etc. *Protocol*: While these efforts used human labelers, they established a rubric that LLM-judges now replicate. Contractors were shown two model responses and asked e.g. “*Which response is more helpful and correct given the user’s request?*”. The “win” choices from thousands of such comparisons trained a reward model. Today, one can prompt a large model with the same task – essentially acting as an automated judge using the RLHF comparison rubric. For instance, an LLM judge might be prompted: “*User asked: ‘How do I bake a cake?’; Assistant1 answered X, Assistant2 answered Y. As a judge following the helpfulness criteria, which answer is better for the user?*”. The judge’s pick (and reasoning) simulates what a human annotator might prefer. **Special instructions** often include definitions of helpfulness or harmlessness and forbid the judge from favoring irrelevant style differences. This category underscores that many LLM-as-a-judge prompts are designed to **mirror human preference rubrics** used in reinforcement learning, but with the speed and consistency of an automated evaluator.

Note: Ranking comparisons are especially favored for *subjective or nuanced evaluations* (e.g. creative writing, conversational helpfulness). Research shows that using pairwise comparison can reduce variance and bias – an LLM judge might struggle to reliably give an absolute “8/10” vs “9/10”, but can more consistently say “answer A is better than B” ³⁰. However, pairwise judging requires more evaluations ($O(n^2)$ comparisons for n items). Thus, it’s often used in benchmarking top models or for fine-tuning reward models, whereas point scores might be used when a quick rating is needed for a single output.

Sources: The above rubric types and examples are drawn from recent peer-reviewed literature and benchmarks. Key references include the TruthfulQA benchmark for binary truthfulness ¹, FlowAI’s evaluator description for scoring scales ² ³, summarization and dialog evaluation studies using Likert scales ⁵, OpenAI and Microsoft research on GPT-4 evaluators (G-Eval) ⁷ ⁹ and GPTScore ³³, fine-grained evaluation frameworks like FLASK ¹³, scientific QA rubrics ¹⁵, the RUBICON conversation evaluation method ¹¹ ²², and LLM preference comparison studies from 2023 ²⁷ ²⁸. These works

collectively show how **LLM-as-a-judge** has evolved a variety of rubric styles to approximate human evaluation across different open-ended tasks (from QA and summarization to coding and dialogue). Each rubric type has its own strengths and use cases, and ongoing research (2024–2025) continues to refine these evaluation protocols for better reliability and alignment with human judgments.

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